A Systematic Comparison of Dimensionality Reduction for Multi-Class Sleep Disorder Classification

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***Abstract*—This study proposes an evaluation framework to optimize multi-class sleep disorder classification (None, Sleep Apnea, Insomnia) using physiological and lifestyle data. To balance diagnostic accuracy and computational cost, we evaluate three experimental scenarios: (1) feature selection using Mutual Information (MI); (2) independent feature extraction (PCA, LDA, ICA, UMAP); and (3) a two-stage hybrid pipeline. The results show that the hybrid method (*E3: Top-4 PCA-RF*) achieves the highest nominal accuracy of $91.99\% \pm 6.10\%$. However, statistical testing indicates no significant performance differences ($p > 0.05$) among the optimal configurations of all three strategies, with the single-stage selection (*E1: Top-4 RF*) reaching 91.47% and the dimensionality reduction configuration (*E2: ICA-SVC*) achieving 91.19%. Given this statistical equivalence, model selection is determined by empirical hardware constraints for edge deployment. Resource benchmarking reveals that the linear framework using ICA combined with a Support Vector Classifier (*E2: Full ICA-SVC (13D)*) requires a minimal training time of 22 ms and an inference latency of just 72 μs/sample, cutting computational overhead substantially compared to the other approaches. Therefore, this study concludes that the *E2: ICA-SVC* configuration is the most optimal strategy, minimizing power and processing footprints while maintaining high diagnostic reliability, making it highly practical for deployment onto smart wearable systems.**

***Index Terms*—Dimensionality Reduction, Feature Selection, Multi-Class Classification, Sleep Disorder.**

I. Introduction

Sleep is essential for physical restoration and cognitive functioning, and disrupted sleep is increasingly recognized as a major public-health concern [1]. Obstructive sleep apnea (OSA) is among the most impactful sleep disorders due to its high prevalence and links to excessive daytime sleepiness and cardiometabolic complications [2]. Despite this burden, scalable identification remains difficult: polysomnography is accurate but resource-intensive, whereas questionnaire-based screening is convenient yet inherently subjective [3], [4].

The growing availability of lifestyle and basic health measurements has enabled supervised learning approaches that infer sleep-related conditions from routinely collected indicators [5]. This study formulates sleep disorder prediction as a multi-class classification problem using demographic, behavioral, and physiological variables [7]. To enhance predictive clarity and computational efficiency, we compare three distinct experimental setups: Experiment 1 applies Mutual Information (MI) [6] based feature selection alone, Experiment 2 applies dimensionality reduction across the full feature set, and Experiment 3 integrates MI-based feature selection as a precursor to reduction. Multiple reduction families (PCA, LDA, ICA, UMAP) and representative classifiers (LR, SVC, RF, AdaBoost, LightGBM, KNN) are assessed under these paradigms. Performance is analyzed as a function of the reduced dimensionality to identify whether pre-optimizing the feature space through MI-based selection provides a superior foundation for multi-class sleep disorder prediction.

II. Related work

The Sleep Health and Lifestyle dataset [7] has been increasingly adopted to study sleep health using mixed-type demographic, behavioral, and basic physiological variables [8]–[10]; however, most existing work is oriented toward sleep-quality analysis rather than multi-class sleep disorder prediction. In particular, regression and statistics-driven studies quantify associations between sleep outcomes and factors such as physical activity, daily steps, stress level, heart rate variability, and blood-pressure-related indicators [9], [11], providing explanatory evidence that these variables are informative for sleep-health analytics [8], [10].

In contrast, limited literature provides actionable guidance on feature representation design for disorder classification on this dataset. A representative machine-learning study benchmarks multiple classifiers and reports strong results for ensemble methods under its setting, yet it formulates the target as sleep quality treated as a categorical label, which differs from predicting clinically meaningful sleep disorder categories [10]. Consequently, comparative evidence that isolates dimensionality reduction as a controlled design factor—across linear projections (PCA) [12], supervised discriminative projections (LDA) [13], independence-based methods (ICA) [14], and non-linear manifold learning (UMAP) [15]—remains limited for sleep disorder prediction on mixed-type lifestyle data.

Beyond dataset-specific studies, dimensionality reduction has been explored in sleep-related machine learning and broader biomedical pipelines, indicating that representation choice can materially affect downstream performance and may interact with class-imbalance handling [16], [17]. Based on these observations, this paper conducts a controlled comparative evaluation of multiple dimensionality-reduction families and assesses their impact on multi-class sleep disorder prediction using six classifiers (Logistic Regression, SVC, Random Forest, AdaBoost, LightGBM, and KNN). Performance is analyzed across varying reduced dimensionalities to characterize the performance-dimensionality relationship and to identify compact representations that maintain or improve classification accuracy.

III. Exploratory Data Analysis

This study uses the public *Sleep Health and Lifestyle* dataset [7], which contains records from 374 individuals. The prediction target is the *Sleep Disorder* label with three classes: *None*, *Sleep Apnea*, and *Insomnia*. The label distribution is imbalanced, where the *None* class is dominant: *None* has 219 samples (58.56%), *Sleep Apnea* has 78 samples (20.86%), and *Insomnia* has 77 samples (20.59%). This class imbalance motivates imbalance-aware training strategies in later stages.

The dataset contains 0 missing values. In terms of feature types, there are 9 numerical variables (1 float and 8 integers) and 3 categorical variables, with *Sleep Disorder* as the categorical target. The numerical features include *Age*, *Sleep Duration*, *Quality of Sleep*, *Physical Activity Level*, *Stress Level*, *Heart Rate*, *Daily Steps*, *Systolic_BP*, and *Diastolic_BP*. The categorical features are *Gender* (binary categorical), *Occupation* (multi-category), and *BMI Category* (multi-category).

IV. Methodology

A. Data Preprocessing

Fig. 1 summarizes the end-to-end workflow of this study, from data exploration to model evaluation. This section details the preprocessing steps used to construct the learning-ready feature matrix.

Data Preparation and Balancing: The *Blood Pressure* attribute was decomposed into its systolic and diastolic components, while *PersonID* was excluded. Categorical variables were processed using One-Hot Encoding, resulting in a 26-dimensional feature space. Following standardization of continuous attributes, the Synthetic Minority Over-sampling Technique (SMOTE) [18]–[20] was applied to address class imbalance. Crucially, over-sampling was restricted to the training partition to ensure an objective evaluation of model generalization on the original test distribution.

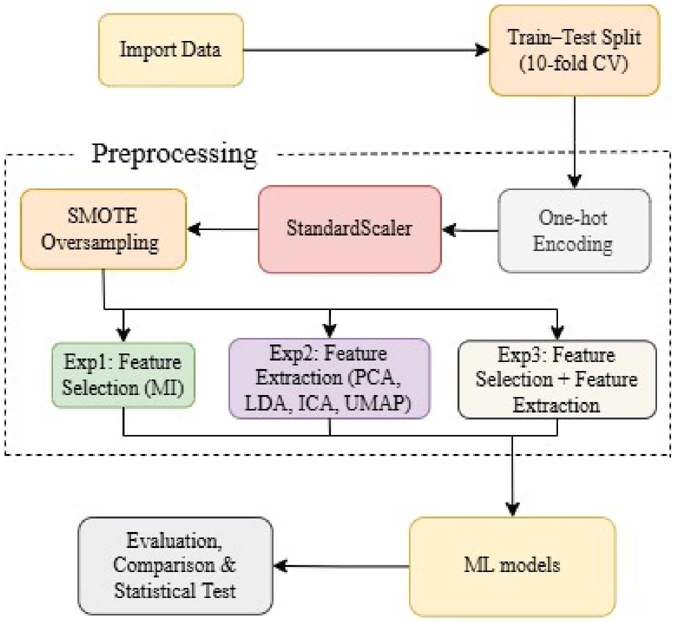

Fig. 1: End-to-End Evaluation Framework Including the Three-Experiment Comparative Workflow.

B. Experimental Design

We evaluate the dimensionality reduction techniques through a three-stage comparative study designed to identify the optimal trade-off between representation complexity and predictive accuracy [21].

1) Experiment 1: Optimization using Mutual Information: To make our models more efficient and find the most useful data, we use a backward feature elimination method based on Mutual Information (MI) scores. First, we calculate the MI score for all 12 features. This score shows how much each feature relates to our target variable, allowing us to rank them from the most to the least important.

Next, we evaluate the six classification models: Logistic Regression, SVC, Random Forest, AdaBoost, LightGBM, and KNN while removing the lowest-ranked features one by one. By checking the model's accuracy at each removal step, we can clearly see when dropping unhelpful features leads to the best results.

2) Experiment 2: Dimensionality Reduction: This stage utilizes the full 26-dimensional feature set across six classifiers: Logistic Regression, SVC, Random Forest, AdaBoost, LightGBM, and KNN [22], [23]. For each reduction algorithm (PCA, ICA, LDA, UMAP), the number of components (n) is chosen from minimum to maximum number to determine the highest performance configuration gains.

3) Experiment 3: MI-Based Optimization + Dimensionality Reduction: This experiment explores a combined approach: using Mutual Information (MI) for feature selection, followed by dimensionality reduction. First, we rank all 12 features based on their MI scores (Figure 2). We then evaluate the models using the top n features, with n ranging from 1 to 8. The goal of this step is to identify the best-performing model for each subset. Finally, once the best model with highest accuracy and its optimal feature subset are identified, we apply dimensionality reduction algorithms (PCA, LDA, ICA, and UMAP) to this specific subset. We evaluate the model's accuracy on these reduced spaces to determine if combining MI-based selection with dimensionality reduction leads to better performance.

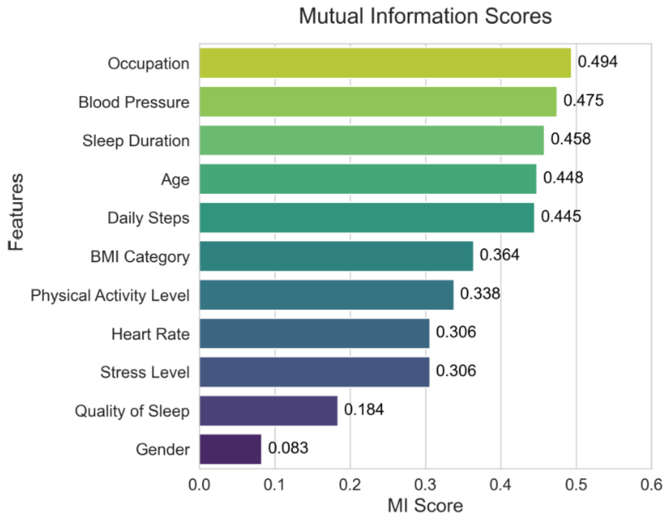

Fig. 2: Mutual Information Scores for Feature Selection Analysis.

V. Results and Discussion

A. Analysis of Experiment 1: Feature Selection via Mutual Information

The results from Experiment 1 illustrate the application of Mutual Information (MI) for feature space reduction. As shown in the performance chart (Figure 3), the relationship between the number of retained Top-N features and model accuracy exhibits a non-linear trend. Initially, isolating the highest-ranked features helps reduce data noise, allowing the models to maintain or slightly improve their classification performance. Conversely, retaining too few attributes (particularly below Top-2) leads to a decrease in accuracy, likely due to the loss of predictive signals.

Specifically, as shown in TABLE I, which summarizes the best-performing Top-K subset of each classifier together with its precision, recall, and F1-score, the Random Forest model observes a favorable balance when utilizing the Top-4 MI-selected features, recording an accuracy of $91.47\% \pm 6.62\%$, a precision of $89.21\% \pm 6.80\%$, a recall of $90.02\% \pm 8.86\%$, and an F1-score of $89.14\% \pm 7.79\%$. This trend indicates that MI-based static feature selection offers a viable trade-off between model simplicity and diagnostic capability. It suggests that a selected subset can maintain adequate performance while requiring less computational effort compared to processing the entire dataset.

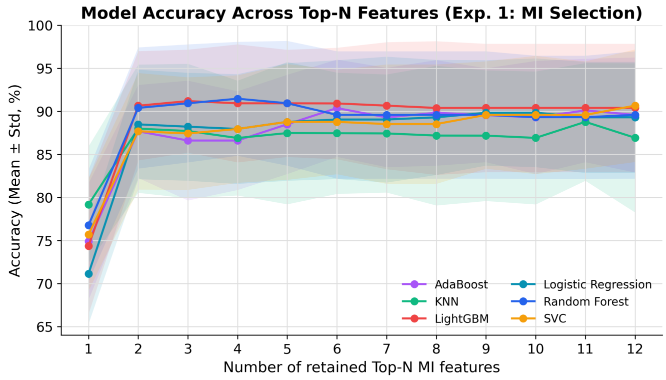

Fig. 3: The Relationship Between the Number of Retained Top-N MI Features and Model Accuracy.

B. Analysis of Experiment 2: Dimensionality Reduction

Experiment 2 evaluates the direct application of dimensionality reduction techniques on the full 26-feature space. As TABLE II shows, linear reduction methods generally provide stable outcomes. In this setup, the ICA-SVC model achieves the highest accuracy, reaching $91.19\% \pm 6.74\%$ with $n = 13$ components. PCA-based configurations also show consistent performance; for example, Logistic Regression reaches $89.84\% \pm 5.95\%$ at $n = 16$. These results suggest that although the original dataset contains a large amount of discriminative information, linear extraction methods tend to require a high number of components to maintain accuracy.

In contrast, the non-linear manifold learning method (UMAP) does not show a performance advantage over linear methods. The highest accuracy observed in the UMAP group in this experiment (such as UMAP-KNN at $n = 7$) only reaches 88.54%, which is lower than the linear models. This finding indicates that applying complex non-linear spatial transformations may not provide additional useful diagnostic information for this dataset.

C. Analysis of Experiment 3: Hybrid MI-Based Optimization and Dimensionality Reduction

The empirical results within Experiment 3 identify the optimal configuration when coupling dual-stage feature selection and latent space extraction. As delineated in TABLE III, the hybrid pipeline integrating the Top-4 MI-selected features, a PCA layer retaining 8 components, and a Random Forest classifier (*E3: Top-4 PCA-RF (8D)*) establishes the peak nominal accuracy of $91.99\% \pm 6.10\%$. This is consistently mirrored by balanced complementary metrics, including a Precision of $89.79\% \pm 6.26\%$, a Recall of $90.69\% \pm 8.21\%$, and an F1-score of $89.85\% \pm 7.10\%$.

This synergy indicates that when the feature space is condensed by the initial information-theoretic layer, the linear projection of PCA remains mathematically robust enough to construct an 8-dimensional latent domain with sufficient

TABLE I: Experiment 1: Best Top-K MI Subset Per Classifier.

Classifier	**Top-K**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 (%)**
Random Forest	**4**	91.47 ± 6.62	89.21 ± 6.80	90.02 ± 8.86	89.14 ± 7.79
LightGBM	3	91.19 ± 5.98	88.91 ± 6.13	89.24 ± 8.13	88.62 ± 6.98
SVC	12	90.66 ± 6.54	88.30 ± 6.34	88.79 ± 8.37	87.92 ± 7.32
AdaBoost	6	90.38 ± 5.65	87.61 ± 6.47	88.20 ± 8.04	87.42 ± 7.02
Logistic Regression	10	89.84 ± 6.31	88.06 ± 6.14	88.59 ± 8.07	87.27 ± 7.12
KNN	11	88.78 ± 6.83	86.17 ± 6.49	86.98 ± 8.64	85.62 ± 7.75

discriminative variance for decision-tree ensembles. Correspondingly, configurations incorporating the non-linear UMAP algorithm remain entirely absent from the upper tier of the performance matrix. The systematic exclusion of these complex geometric projection structures implies that adding complex non-linear spatial transformations yields no tangible structural benefits for advancing the baseline classification boundaries.

D. Linear vs. Non-linear Comparison

A central aspect of the analysis is the comparison between linear methods (PCA, LDA, ICA) and the non-linear manifold learning method (UMAP). Across all experiments, linear configurations consistently maintain highly competitive nominal performance. In Experiment 2, the linear model (ICA-SVC) marginally outperforms the best representative of the non-linear method (UMAP-KNN). Furthermore, in the overall ranking of optimal configurations, combinations based on linear features demonstrate significant prominence.

It is worth noting that linear methods achieve this level of performance without requiring the extensive computational overhead associated with distance metrics and neighborhood graph constructions inherent to UMAP. This performance characteristic implies that the physiological and demographic attributes in this specific representation interact robustly with standard linear models. Consequently, the additional complexity introduced by non-linear manifold mapping may not offer a clear empirical advantage for this feature space structure.

E. Statistical Verification

To ensure objectivity in model selection, the study used the Corrected Paired Student's t-test with a significance level of $\alpha = 0.05$ [24], [25]. The null hypothesis (H_0) assumes that there is no significant difference in classification performance between the best linear models (or the static MI filtering model) and the non-linear models (UMAP).

The extensive cross-checking results have reinforced this observation. In Experiment 2, the test between the best linear configuration (ICA-SVC, $n = 13$) and the corresponding non-linear configuration (UMAP-KNN, $n = 7$) returned a value of $p = 0.21125$. Similarly, in Experiment 3, the comparison between the PCA-based hybrid model (Top-4 PCA-Random Forest, 8D) and the UMAP-based hybrid model (Top-3 UMAP-Random Forest, 9D) yielded a value of $p = 0.12336$. Notably, even when comparing the simplest configuration of Experiment 1 (MI Random Forest, Top-4) with the complex UMAP configuration (UMAP-KNN, $n = 7$), the results still show equivalence with $p = 0.10775$.

Furthermore, we compare the optimal model of each experiment against one another to verify that no single strategy holds a statistically superior position. The hybrid model of Experiment 3 (Top-4 PCA-Random Forest, 91.99%) shows no significant difference from the Experiment 1 selection model (MI Random Forest, Top-4, 91.47%, $p = 0.3288$), nor from the Experiment 2 reduction model (ICA-SVC, 91.19%, $p = 0.3567$); likewise, the Experiment 1 and Experiment 2 optima are statistically indistinguishable ($p = 0.7053$).

Because $p > 0.05$ in all comparison scenarios, there is not enough evidence to reject the H_0 hypothesis. This statistical evidence confirms that linear configurations (or using original features through MI filtering) provide statistically equivalent accuracy compared to non-linear manifold learning. This analysis plays a key role in drawing the conclusion: linear configurations or models optimized through MI should be prioritized. They achieve an equal level of performance but are far superior in interpretability, compactness, and the ability to save computational costs, matching the criteria for application on resource-constrained diagnostic devices.

F. Computational Efficiency Benchmark

We benchmark the wall-clock training time and per-sample inference latency of the best configuration of each strategy under one identical protocol (10-fold cross-validation, 10 repeats). Fig. 4 presents the accuracy–runtime trade-off, while TABLE IV reports the measured times in plain units for readability. In brief, the linear pipelines remain real-time (sub-millisecond inference), whereas the non-linear UMAP layer is markedly more expensive in both training and inference.

G. Discussion

The experimental results and statistical verification suggest two findings regarding the characteristics of the sleep disorder diagnosis data and the effectiveness of model optimization strategies.

1) Structural Separability of the Data: Linear projections matching non-linear manifold learning on this dataset suggests that the three classes (None, Insomnia, Sleep Apnea) are separable by standard classifiers without high-dimensional non-linear expansion. When performance is comparable, linear methods are preferred for their determinism, interpretability, and lower computational cost at both training and inference.

TABLE II: Performance Comparison for Experiment 2 (Dimensionality Reduction, 10-fold CV).

Reduction		**Metric**	**KNN**	**LR**	**SVC**	**RF**	**AdaBoost**	**LGBM**
Linear	PCA	Optimal n	16	16	16	14	17	13
		Accuracy (%)	88.26 ± 6.94	89.84 ± 5.95	90.66 ± 6.54	89.84 ± 6.43	90.39 ± 6.45	90.92 ± 6.99
		Precision (%)	85.50 ± 6.53	87.65 ± 6.07	88.30 ± 6.34	87.11 ± 7.72	88.03 ± 6.25	88.59 ± 7.39
		Recall (%)	86.37 ± 8.41	88.50 ± 7.66	88.79 ± 8.37	89.53 ± 8.19	88.96 ± 8.61	89.71 ± 9.02
		F1 (%)	85.00 ± 7.70	87.41 ± 6.69	87.92 ± 7.32	87.58 ± 7.51	87.79 ± 7.51	88.64 ± 8.08
	LDA	Optimal n	2	2	2	2	2	2
		Accuracy (%)	89.05 ± 7.31	89.05 ± 6.77	90.12 ± 6.82	89.04 ± 6.40	87.44 ± 6.23	88.24 ± 6.48
		Precision (%)	86.58 ± 6.81	86.80 ± 6.30	87.86 ± 6.08	86.50 ± 7.57	84.74 ± 6.20	85.65 ± 7.57
		Recall (%)	88.40 ± 8.48	88.42 ± 7.95	88.77 ± 8.25	88.87 ± 7.87	87.86 ± 7.96	88.43 ± 7.84
		F1 (%)	86.83 ± 7.68	86.91 ± 7.16	87.75 ± 7.28	86.87 ± 7.43	85.16 ± 6.82	86.11 ± 7.50
	ICA	Optimal n	3	13	13	9	16	16
		Accuracy (%)	88.78 ± 6.30	89.84 ± 6.31	91.19 ± 6.74	89.59 ± 7.30	89.30 ± 6.42	91.19 ± 6.56
		Precision (%)	85.63 ± 6.61	87.57 ± 6.20	89.33 ± 6.19	86.82 ± 8.62	87.40 ± 6.78	88.98 ± 7.00
		Recall (%)	87.05 ± 7.95	89.04 ± 7.67	89.84 ± 8.18	89.07 ± 9.14	88.21 ± 8.22	89.89 ± 8.28
		F1 (%)	85.66 ± 7.03	87.69 ± 6.76	88.95 ± 7.28	87.29 ± 8.55	86.76 ± 7.24	89.08 ± 7.45
Non-Linear Learning	UMAP	Optimal n	7	19	13	26	23	8
		Accuracy (%)	88.54 ± 8.71	87.75 ± 8.38	88.54 ± 8.37	88.27 ± 7.62	86.95 ± 9.13	86.69 ± 9.06
		Precision (%)	86.43 ± 8.50	85.35 ± 8.27	86.31 ± 8.03	85.42 ± 7.97	84.21 ± 9.29	84.46 ± 8.77
		Recall (%)	87.74 ± 10.13	87.43 ± 9.84	87.88 ± 9.88	87.74 ± 9.41	86.77 ± 10.14	84.43 ± 9.29
		F1 (%)	86.33 ± 9.47	85.51 ± 9.06	86.42 ± 9.08	85.80 ± 8.57	84.53 ± 9.79	83.74 ± 9.30

TABLE III: Performance Ranking of Optimal Configurations Across Experiments.

Rank	**Configuration**	**Accuracy (%)**	**Precision (%)**	**Recall (%)**	**F1 (%)**
1	**E3: Top-4 PCA-Random Forest (8D)**	91.99 ± 6.10	89.79 ± 6.26	90.69 ± 8.21	89.85 ± 7.10
2	E3: Top-3 ICA-LightGBM (9D)	91.72 ± 5.92	89.59 ± 6.09	90.21 ± 7.87	89.53 ± 6.83
3	E3: Top-5 ICA-Random Forest (9D)	91.47 ± 6.92	89.32 ± 6.96	90.18 ± 8.87	89.25 ± 7.96
4	E1: Top-4 RF (MI only)	91.47 ± 6.62	89.21 ± 6.80	90.02 ± 8.86	89.14 ± 7.79
5	E2: Full ICA-SVC (13D)	91.19 ± 6.74	89.33 ± 6.19	89.84 ± 8.18	88.95 ± 7.28
6	E3: Top-6 ICA-LightGBM (9D)	91.19 ± 6.68	88.81 ± 7.07	90.22 ± 8.52	89.12 ± 7.58
7	E3: Top-7 ICA-LightGBM (11D)	91.19 ± 6.68	88.85 ± 7.12	89.97 ± 8.75	88.94 ± 7.76
8	E3: Top-2 ICA-Random Forest (4D)	91.19 ± 5.97	89.49 ± 5.83	89.86 ± 7.91	89.22 ± 6.66
9	E3: Top-8 PCA-LightGBM (19D)	90.66 ± 7.95	88.18 ± 8.71	89.05 ± 10.48	88.08 ± 9.60
10	E3: Top-1 PCA-KNN (1D)	79.17 ± 6.76	77.41 ± 9.20	71.20 ± 7.32	71.50 ± 7.48

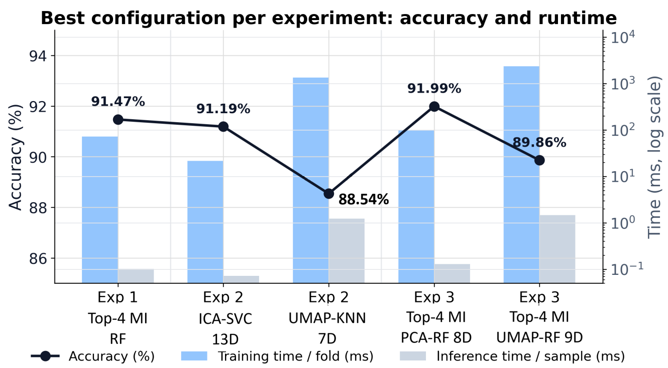

Fig. 4: Best Configuration Per Strategy Documenting Core Accuracy Metrics Alongside Training and Per-Sample Inference Latencies.

TABLE IV: Training and Per-Sample Inference Time of the Best Configuration Per Strategy (10-Fold $\times$ 10 Repeats).

Configuration	**Dimension**	**Train (ms)**	**Inference (μs/sample)**
E1: Top-4 MI-RF	4	72	102
E2: ICA-SVC	13	22	72
E2: UMAP-KNN	7	1345	1239
E3: Top-4 MI-PCA-RF	8	99	129
E3: Top-4 MI-UMAP-RF	9	2357	1468

2) *Engineering Trade-offs and Optimal Deployment Strategy:* A practical insight derived from this benchmark lies in the intersection of statistical equivalence and computational runtime indicators. Although the hybrid optimization pipeline in Experiment 3 (*E3: Top-4 PCA-RF*) records the highest nominal classification accuracy (91.99%), our corrected paired t-test matrix confirms the absence of statistically significant

performance differences ($p > 0.05$) among the optimal configurations across all three strategies. Specifically, the leading model of Experiment 3 exhibits strict statistical equivalence when cross-compared against the selection pipeline of Experiment 1 (91.47%, $p = 0.3288$) and the reduction pipeline of Experiment 2 (91.19%, $p = 0.3567$).

When diagnostic reliability remains statistically indistinguishable across strategies, the selection of a configuration for physical integration is determined by empirical hardware execution metrics, specifically training duration and forward-pass inference latency. Under this engineering paradigm, the linear feature extraction framework utilizing Independent Component Analysis combined with a Support Vector Classifier (*E2: Full ICA-SVC (13D)*) demonstrates strong viability for deployment within resource-constrained hardware environments.

According to the quantitative resource profiles documented in TABLE IV and Fig. 4, the *E2: ICA-SVC* pipeline requires 22 ms for wall-clock training, representing a reduction in computational duration of approximately 69.4% compared to *E1: Top-4 RF* (72 ms) and 77.8% compared to the hybrid *E3: PCA-RF* (99 ms). More importantly, for continuous, real-time on-chip screening tasks on wearable microcontrollers, the *E2: ICA-SVC* framework records a per-sample inference latency of only 72 μs. This operational duration is shorter than the 102 μs required by the plain MI protocol and the 129 μs demanded by the multi-stage hybrid structure.

By utilizing a low-complexity linear projection matrix to map the features into an independent 13-dimensional domain, this configuration minimizes both processing footprint and power consumption profiles while maintaining diagnostic reliability equivalent to more complex alternatives. Consequently, the *E2: Full ICA-SVC* architecture serves as a practical choice for deployment onto smart wearable healthcare devices characterized by strict hardware and battery limitations.

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REFERENCES

[1] N. Baranwal, P. K. Yu, and N. S. Siegel, "Sleep physiology, pathophysiology, and sleep hygiene," *Prog. Cardiovasc. Dis.*, vol. 77, pp. 59-69, Mar.-Apr. 2023, doi: 10.1016/j.pcad.2023.02.005.

[2] Y. Yeghiazarians *et al.*, "Obstructive sleep apnea and cardiovascular disease: A scientific statement from the American Heart Association," *Circulation*, vol. 144, no. 3, pp. e56-e67, Jul. 2021, doi: 10.1161/CIR.0000000000000988.

[3] A. Abrishami, A. Khajehdehi, and F. Chung, "A systematic review of screening questionnaires for obstructive sleep apnea," *Can. J. Anaesth.*, vol. 57, no. 5, pp. 423-438, May 2010, doi: 10.1007/s12630-010-9280-x.

[4] O. Hussein, A. Alkhader, A. Gohar, and A. Bhat, "Home Sleep Apnea Testing for Obstructive Sleep Apnea," *Mo. Med.*, vol. 121, no. 1, pp. 60-65, Jan.-Feb. 2024.

[5] S. Aziz et al., "Wearable Artificial Intelligence for Sleep Disorders: Scoping Review," *J. Med. Internet Res.*, vol. 27, e65272, May 2025, doi: 10.2196/65272.

[6] H. Peng, F. Long, and C. Ding, "Feature selection based on mutual information: criteria of max-dependency, max-relevance, and min-redundancy," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 27, no. 8, pp. 1226-1238, Aug. 2005, doi: 10.1109/TPAMI.2005.159.

[7] L. Tharmalingam, "Sleep Health and Lifestyle Dataset," Kaggle, Sep. 18, 2023. [Online]. Available: https://www.kaggle.com/datasets/uom190346a/sleep-health-and-lifestyle-dataset.

[8] R. Khanra, S. Guria Das, A. Samanta, S. Sarkar, A. K. Sinha, and S. Mukherjee, "Association between sleep quality and depressive symptoms among adults: analysis of the Sleep Health and Lifestyle dataset," *International Journal of Progressive Research in Engineering Management and Science (IJPREMS)*, vol. 5, no. 12, pp. 853-861, Dec. 2025, doi: 10.58257/IJPREMS45500.

[9] X. Ma and Z. Wang, "Research on the relationship between lifestyle and sleep health," in *Highlights in Science, Engineering and Technology: CMLAI 2024*, vol. 94, pp. 379-385, 2024. [Online]. Available: https://drpress.org/ojs/index.php/HSET/article/view/20612/20176.

[10] M. U. Bhatti, A. Saeed, M. Ashir, N. Hussain, M. Anwar, and M. Farhat Ullah, "Modeling sleep health and lifestyle using supervised learning algorithms," *Journal of Computing & Biomedical Informatics*, vol. 9, no. 1, 2025, Art. no. 998-0901/2025, doi: 10.56979/901/2025.

[11] H. ChuDuc, K. NguyenPhan, and D. NguyenViet, "A review of heart rate variability and its applications," *APCBEE Procedia*, vol. 7, pp. 80-85, 2013, doi: 10.1016/j.apcbee.2013.08.016.

[12] I. T. Jolliffe and J. Cadima, "Principal component analysis: a review and recent developments," *Philos. Trans. R. Soc. A*, vol. 374, no. 2065, p. 20150202, Apr. 2016, doi: 10.1098/rsta.2015.0202.

[13] A. Tharwat, T. Gaber, A. Ibrahim, and A. E. Hassanien, "Linear discriminant analysis: A detailed tutorial," *AI Commun.*, vol. 30, no. 2, pp. 169-190, 2017, doi: 10.3233/AIC-170729.

[14] A. Hyvärinen and E. Oja, "Independent component analysis: algorithms and applications," *Neural Netw.*, vol. 13, no. 4-5, pp. 411-430, May-Jun. 2000, doi: 10.1016/S0893-6080(00)00026-5.

[15] L. McInnes, J. Healy, and J. Melville, "UMAP: Uniform Manifold Approximation and Projection for Dimension Reduction," *J. Open Source Softw.*, vol. 3, no. 29, p. 861, 2018, doi: 10.21105/joss.00861.

[16] L. Biedebach *et al.*, "Unsupervised machine learning in sleep research: a scoping review," *Sleep*, vol. 48, no. 11, p. zsaf189, Nov. 2025, doi: 10.1093/sleep/zsaf189.

[17] A. Jamil, M. K. Hanif, M. U. Sarwar, and M. IˆKhan, "An empirical evaluation of dimensionality reduction and class balancing for medical text classification," *Sci. Rep.*, 2025, doi: 10.1038/s41598-025-30537-w.

[18] H. He and E. A. Garcia, "Learning from imbalanced data," *IEEE Trans. Knowl. Data Eng.*, vol. 21, no. 9, pp. 1263-1284, Sep. 2009, doi: 10.1109/TKDE.2008.239.

[19] A. S. Tarawneh, A. B. Hassanat, G. A. Altarawneh, and A. Almuhaimeed, "Stop Oversampling for Class Imbalance Learning: A Review," *IEEE Access*, vol. 10, pp. 47643-47660, 2022, doi: 10.1109/ACCESS.2022.3169512.

[20] A. Fernández, S. García, F. Herrera, and N. V. Chawla, "SMOTE for Learning from Imbalanced Data: Progress and Challenges, Marking the 15-Year Anniversary," *J. Artif. Intell. Res.*, vol. 61, pp. 863-905, Apr. 2018, doi: 10.1613/jair.1.11192.

[21] A. A. Wani, "Comprehensive review of dimensionality reduction algorithms: challenges, limitations, and innovative solutions," *PeerJ Comput. Sci.*, vol. 11, p. e3025, Jul. 2025, doi: 10.7717/peerj-cs.3025.

[22] G. Ke *et al.*, "LightGBM: A Highly Efficient Gradient Boosting Decision Tree," in *Proc. Adv. Neural Inf. Process. Syst. (NeurIPS)*, 2017, pp. 3146-3154.

[23] J. Cervantes, F. García-Lamont, L. Rodríguez-Mazahua, and A. Lopez, "A comprehensive survey on support vector machine classification: Applications, challenges and trends," *Neurocomputing*, vol. 408, pp. 189-215, Sep. 2020, doi: 10.1016/j.neucom.2019.10.118.

[24] O. Rainio, J. Teuho, and R. Klén, "Evaluation metrics and statistical tests for machine learning," *Sci. Rep.*, vol. 14, Art. no. 6086, Mar. 2024, doi: 10.1038/s41598-024-56706-x.

[25] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Mach. Learn.*, vol. 52, no. 3, pp. 239-281, 2003, doi: 10.1023/A:1024068626366.